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ARPA: Augmented Reality Prism Adaptation induces sensorimotor and visuospatial effects

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ABSTRACT

Prism adaptation (PA) is a well-established method for sensorimotor recalibration and influencing visuospatial processing. It is also one of the rehabilitation approaches for neglect patients. Recent studies have shown effective adaptation in virtual reality (VR) settings simulating the classic PA procedure. However, no research has explored prism adaptation in augmented reality (AR), which combines the advantages of VR with greater ecological validity, allowing individuals to perform a virtual PA procedure in a real environment with natural visual feedback from their own hand. The present study introduces **Augmented Reality Prism Adaptation (ARPA)**, a novel procedure that incorporates the benefits of AR with the classic PA technique. **Forty-eight healthy participants** underwent either leftward or rightward ARPA, and their sensorimotor and visuospatial aftereffects were evaluated immediately post-ARPA and over a 40-minute period. Results revealed significant and long-lasting sensorimotor aftereffects following both leftward and rightward ARPA, while only leftward ARPA induced an immediate rightward visuospatial aftereffect. Importantly, we investigated the generalization of these changes to the real environment, finding that ARPA-induced sensorimotor aftereffects extended beyond the virtual setting. While these findings show that ARPA produces both sensorimotor and visuospatial aftereffects, further research is essential to evaluate its applicability and effectiveness in neglect rehabilitation.

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Introduction

Prism adaptation (PA) is a well-established method that reversibly modifies sensorimotor correspondences by inducing visuomotor recalibration, a process that corrects sensorimotor misalignments caused by visual shifts and allows individuals to adjust their motor actions accordingly (Facchin et al., 2019; Helmholtz, 1867; Redding et al., 2005; Redding & Wallace, 1996; Redding & Wallace, 2019; Stratton, 1896; Wilkinson, 1971). Adaptation to laterally displaced vision has been shown to affect not only sensorimotor correspondences, resulting in motor adjustments to altered visual inputs, but also higher-order spatial cognition, influencing visuospatial attention and perception. Such effects on spatial cognition have been observed in both healthy and brain-damaged individuals, depending on the direction of the visual displacement. On the one hand, adaptation to rightward deviating prisms is reported to ameliorate the pathological rightward bias exhibited by patients with hemispatial neglect (Frassinetti et al., 2002; Gammeri et al., 2023; Pisella et al., 2006; Rossetti et al., 1998), for whom awareness of the contralesional side of space is lost, usually, following a right hemisphere damage (Corbetta et al., 2005). On the other hand, adaptation to leftward deviating prisms is thought to mimic, albeit with a smaller magnitude, neglect-like behaviour in healthy individuals, modifying visuospatial processing (Berberovic & Mattingley, 2003; Bultitude & Woods, 2010; Colent et al., 2000; Girardi et al., 2004; Goedert et al., 2010; Loftus et al., 2009; Michel et al., 2003; Reed & Dassonville, 2014), as well as temporal processing (Anelli & Frassinetti, 2019; Casadio, Patané, Ballotta, et al., 2024; Casadio, Patané, Candini, et al., 2024; Frassinetti et al., 2009), and even reward-based learning (Schintu et al., 2018). The changes induced by PA not only depend on the direction of the visual displacement, but also the state of the system: intact versus damaged brain (for comprehensive reviews on the different neural mechanisms of prism adaptation in healthy adults and individuals with spatial neglect, please see Boukrina & Chen, 2021; Panico et al., 2020). In neglect patients, while right PA has been reported to improve neglect symptoms (Rossetti et al., 1998; Serino et al., 2009) left PA neither worsens their rightward bias nor affects their sensorimotor behaviour (Luaute et al., 2012). In healthy individuals, instead, while both left and right PA modulate sensorimotor performance, only left PA modulates visuospatial performance (Colent et al., 2000; Michel et al., 2003; Michel & Cruz, 2015; Patané et al., 2016; Schintu et al., 2014, 2018) with few exceptions (Berberovic & Mattingley, 2003; McIntosh et al., 2023). Since most of the studies showing a significant visuospatial modulation following left PA report no behavioral modulation after right PA, the latter is typically used as control for the left PA aftereffects (Bultitude et al., 2013; Colent et al., 2000; Loftus et al., 2009; Michel & Cruz, 2015; Patané et al., 2016; Reed & Dassonville, 2014; Schintu et al., 2014, 2018, 2020, 2022). Another critical difference between sensorimotor and visuospatial aftereffects concerns their time

course: while sensorimotor aftereffects are observed immediately after PA and significantly decrease over time, visuospatial aftereffects can be observed immediately after PA, although they take some time to develop fully and then fluctuate over about 40 min following adaptation (Schintu et al., 2014).

PA is still extensively investigated in both healthy individuals and neglect patients, since neglect rehabilitation remains challenging (Meidian et al., 2022). Among the available approaches, PA has been considered the most feasible intervention (Barrett & Houston, 2019), highlighting its advantages and clinical relevance, when successful, as compared to other rehabilitative approaches (Gammeri et al., 2023). The clinical impact of PA rehabilitation procedures extends to activities of daily life (Anelli & Frassinetti, 2019; Shiraishi et al., 2008), as well as reading (Angeli et al., 2004; Farnè et al., 2002) and wheelchair driving (Jacquin-Courtois et al., 2008; Watanabe & Amimoto, 2010). Furthermore, interest in PA-induced aftereffects has been revived by novel findings observed in other cognitive functions, such as language (Conte et al., 2024; Turriziani et al., 2021) and in other domains, such as psychiatry (Magnani et al., 2022). However, recent meta-analyses (Li et al., 2021; Qiu et al., 2021) have challenged its clinical efficacy, indicating that the benefits of PA do not consistently translate into meaningful improvements in daily activities, thereby calling for a cautious interpretation of PA therapeutic potential.

In recent years, new technologies have emerged, offering novel possibilities for assessment, diagnosis, treatment, and experimentation (Cavedoni et al., 2022; Ogourtsova et al., 2017; Terruzzi et al., 2023). Virtual reality (VR), which is part of extended reality technologies (Sakai et al., 2020), may allow for a more flexible, controlled, quantifiable, and automatized implementation of sensorimotor adaptation when compared to the classic PA (Bui et al., 2021; Cho et al., 2022; Wilf et al., 2021). Recent studies investigating visuomotor adaptation in VR environments have shown that VR-based PA, similar to a classic PA procedure, induces adaptation in healthy individuals with significant aftereffects.

However, while the sensorimotor aftereffects induced by adaptation in VR settings are consistently reported, even when the employed technology varies significantly (Belger et al., 2024; Cho et al., 2022; Ramos et al., 2019; Wilf et al., 2021), the visuospatial aftereffects induced by virtual PA are inconsistent (Bourgeois et al., 2021; Gammeri et al., 2018; Salatino et al., 2023). For example, Gammeri et al. (2018) found that VR-based adaptation in healthy individuals significantly modulated performance at the **manual line bisection tasks**, but not at the **perceptual version of the task, the so-called landmark task** (Milner et al., 1992). In contrast, in Bourgeois et al. (2021) study, despite the presence of sensorimotor aftereffects immediately after virtual PA, neglect patients did not show significant visuospatial aftereffects, as quantified by **manual line bisection and cancellation tasks** that were performed in the virtual environment. Plausible reasons for these mixed results, which are inconsistent with the literature on canonical PA, could stem from the various VR approaches or different modalities

used to implement PA in virtual settings, compared to the traditional PA conducted in real environments.

The present study aimed at further assessing the visuospatial effects induced by a novel device that simulates PA in augmented reality (AR) and assesses their time course. The main advantage of our novel AR device is that virtual elements are overlaid onto the user's real environment, thus allowing the user to visualize and interact with both real and virtual components simultaneously (Sakai et al., 2020). This AR system simulates PA by having the headset laterally shifting the visual field by 15° rightward or leftward, while participants point toward virtual targets that are embedded into the real physical environment. The device allows receiving natural visual feedback from the user's real hand and movement paths, whose role has been shown to critically modulate PA efficacy in real conditions (Fortis et al., 2010; Redding & Wallace, 1988, 1990). Indeed, at odds with previous work and similar to Cho et al. (2022), here participants received direct visual feedback of their own hand during the pointing task (as they would if they were wearing classic wedge prisms) instead of seeing a virtual avatar handling a controller as a pointing device. Our goal was to test whether Augmented Reality Prism Adaptation (ARPA) would induce both sensorimotor and visuospatial effects, as reported by previously published studies using classic PA procedure. The specific aim of our study was three-fold: a) investigating whether both sensorimotor and visuospatial shifts are induced by ARPA, b) identifying their temporal dynamics, along with the c) generalization/transfer of such aftereffects to the real environment.

To investigate the immediate sensorimotor and the visuospatial aftereffects following a single session of left or right ARPA we used an AR version of the pointing task and the landmark task, respectively (Schintu et al., 2014, 2017). While not a gold standard for the clinical assessment of neglect symptoms, we chose the landmark task due to its sensitivity to lateralized perceptual bias, as it primarily assesses the more perceptual rather than motor component of the line bisection task (Bisiach et al., 1998; Fortis et al., 2011; Milner et al., 1992). This test is often used as a proxy for detecting asymmetry in visuospatial attention, as it quantifies distortions in spatial perception, particularly along the horizontal axis, reflecting leftward or rightward visuospatial biases, in both healthy participants and neglect patient (Harvey & Olk, 2004; McIntosh et al., 2019). Based on PA literature, we expect both left and right ARPA to produce an immediate rightward and leftward sensorimotor shift, respectively, and solely left ARPA is expected to induce an immediate rightward visuospatial shift. To evaluate the time course of the sensorimotor and visuospatial shifts, the AR-pointing task and the landmark task were repeatedly administered for approximately 40 min after a single session of either left or right ARPA. Based on work from our laboratory on PA (Schintu et al., 2014), we expected sensorimotor and visuospatial changes to emerge immediately and last for about 40 min. Both the AR-pointing and the landmark tasks were designated as

outcome measures to assess changes in sensorimotor and visuospatial biases following ARPA, as we aimed to evaluate whether ARPA replicates the effects of traditional PA across both domains. Specifically, the sensorimotor aftereffect induced by right ARPA serves as the primary outcome, allowing for a contrast between left and right ARPA effects. Meanwhile, the visuospatial aftereffect induced by right ARPA acts as a control for the one produced by left ARPA. To assess the possible generalization of the ARPA-induced sensorimotor shifts into real settings, we compared the sensorimotor changes indexed by the AR version of the pointing task to those quantified by the canonical pointing task, which was performed outside the virtual settings. In particular, the comparison between the two versions of sensorimotor tasks was carried out before, right after, and 40 min after a leftward or rightward ARPA. As a result of the generalization of the ARPA effects outside the AR environment, we expected no difference between the sensorimotor changes measured in the two (virtual and real) pointing tasks.

Material and methods

Participants

Forty-eight volunteers with normal or corrected-to-normal vision and no history of neurological or psychiatric disease were recruited for the study. All participants were right-handed according to the Edinburgh Handedness Inventory (Oldfield, 1971), had normal or corrected-to-normal vision (with contact lenses), were compensated for their participation, and gave written informed consent. The study was approved by the ethics committee, Comité de Protection des Personnes Ile de France VI (2018-A03175-50), in accordance with the ethical standards of the 1964 Declaration of Helsinki (2008). The sample size was determined through a priori power analysis using G*Power 3, employing a repeated measures analysis of variance (ANOVA) model. Based on the effect size reported by McIntosh et al. in 2023 (Cohen's $d = 0.27$), we set the parameters as follows: $f = 0.135$, $\alpha = 0.05$, power = 0.80, 9 repeated measures, and two groups, resulting in a projected total sample size of $N = 48$. This sample size estimation also served as a stopping rule, and no additional participants were recruited. Participants were randomly assigned to either the experimental or the control group: 24 participants were adapted to the leftward ARPA (experimental group, 17 females, mean age = 23, SEM = 0.75) and the other 24 to rightward ARPA (control group, 16 females, mean age = 24, SEM = 1.04).

Procedure

Participants underwent the calibration phase of the augmented reality (AR) system so that the VR headset (Vive Pro system®, HTC Europe Co. Ltd.,

Salamanca, Wellington Street, Slough, Berkshire SL1 1YP, U.K.) and the Leap Motion Controller® hand tracking device (Ultraleap, California, USA) provided accurate detection and recording of their hand movements throughout the entire experiment.

As shown in [Figure 1](#), participants underwent a single session of ARPA and several assessment blocks; one before (Pre), one immediately after the adaptation procedure (Post 1), and eight ensuing ones (Post 2–9). In each block, the open-loop pointing task was administered to quantify the sensorimotor performance and the landmark task to measure the visuospatial performance. The block immediately before (Pre) and immediately after adaptation (Post 1) along with the one at the very end (Post 9) consisted of two open-loop pointing blocks, one performed in the real environment and one within the AR environment (i.e., wearing the head-mounted display), which were then followed by the landmark task. The order of the two open-loop tasks for these three blocks was counterbalanced across participants. The first block provided baseline measures of both open-loop pointing and landmark tasks performance, to which the performance of the post-adaptation blocks was compared. To minimize de-adaptation, participants were instructed to keep their eyes closed and avoid moving their right hand in between tasks in the post-adaptation phase. The ARPA session lasted approximately 10–15 min, while each assessment block lasted about 5 min. The complete experiment took approximately 1 h and 15 min.

Augmented reality prism adaption: ARPA

Apparatus

An HTC Vive Pro® system was used to display the virtual stimuli and induce an optical shift during adaptation, with a digital image of 1440×1660 pixels for each eye, allowing a total of 2880×1660 pixels, a visual field of 110° at a refresh rate of 90hz. The head-mounted display can adjust interpupillary distance by

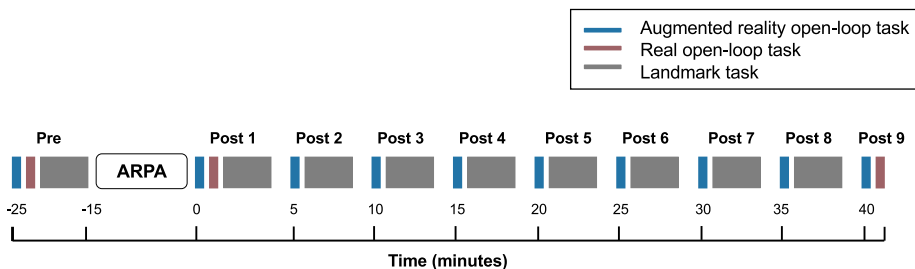


Figure 1. Experimental design and time course of the study. The experiment consisted of ten experimental blocks (one before, one immediately after ARPA and eight ensuing ones) plus a block consisting of open-loop pointing measures after the eighth block. For each of the experimental blocks, open-loop pointing was always performed before the landmark task. For Pre, Post 1 and Post 9, the order of the two open-loop tasks was counterbalanced across participants.

sliding the lenses. The headset and controllers are equipped with inside-out tracking and allow for 6 degrees of freedom head and controller tracking. Those features enable orientation and positional tracking in the virtual space. The real environment was seen through the front camera of the HMD. The hand tracking was handled by a Leap Motion Controller® placed on a rig at 33 cm from the table and angled at 15° and 57 cm from the participant's midline and was controlled via specifically dedicated software (developed by MIMBUS, <https://mimbus.com>) using the game engine UNITY v217.3.03, and ran on a computer with Windows 10, an Intel i5 processor and an NVIDIA GTX 1060 graphic card.

The system was calibrated so that the origin of the three-dimensional AR space was aligned to the participant's midline. To induce adaptation effects, the video feed of the front camera was shifted by a fixed number of degrees in the horizontal direction inside of the HMD, so that the real environment appeared laterally displaced 15° leftward or rightward. The shift was also applied to the virtual elements of the scene. As a result, the ARPA system modified the entire visual field including the real environment, the real hand of the subject and the virtual target.

ARPA procedure

During the adaptation phase, participants were comfortably seated in front of a blackboard with their head positioned on a chinrest while wearing the VR head-mounted display that induced a lateral shift of their visual field by 15° leftward or, as a control, rightward (see [Figure 2](#)). Following verbal instructions given through the Vive headset, participants performed 150 pointing movements either towards the right (+10°), or the left (−10°) virtual target (10 mm diameter) displayed in front of them (*on the black board*; as shown in [Figure 2](#)) in a pseudorandom order. Prior to each pointing movement, participants placed their (unseen) right index finger on the starting position i.e., the chinrest base that was close to and aligned to the midline of their chest. A virtual black rectangle appeared at the bottom of their visual field to prevent participants from seeing their hand during the first third of each pointing movement. Participants were instructed to point with the index finger extended, to execute a one-shot movement at a fast but comfortable speed, and to return their hand to the starting position only when instructed by the experimenter. After the exposure phase, the VR headset was removed by the experimenter and participants were asked to keep their eyes closed and to avoid moving their right hand before starting the assessment blocks. To note that ARPA provided identical tactile feedback compared to traditional PA, as participants in both cases physically would touch the surface of the workspace when completing the reaching movement. In classical PA, participants reach to touch a dot that is physically drawn on the workspace. Similarly, in ARPA, participants interact with the same surface; however, instead of a physically drawn target, a dot is visually projected onto the workspace.

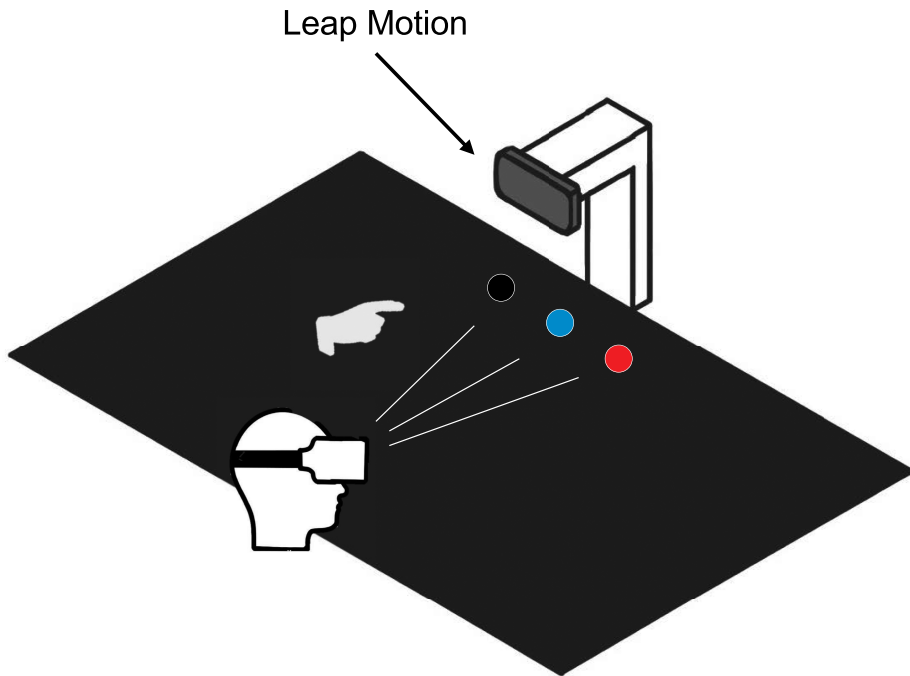


Figure 2. Overview of the ARPA setup. While wearing the head-mounted display, participants can see the targets projected in the augmented reality (AR) environment during the AR prism adaptation procedure.

Open-loop pointing

To assess the sensorimotor shift induced by ARPA we used the open-loop pointing task. Two variants of the open-loop task were administered in this experiment. During the “real” open-loop pointing task participants were seated in front of a board where three targets (5 mm diameter) were positioned at 0° , -10° , and $+10^\circ$, with respect to the participant’s midsagittal plane. Participants were asked to perform six pointing movements towards the central (0°) blue target with their right index. Before each pointing movement, they were verbally instructed to look at the central target, close their eyes, and point to the target at a comfortable speed while keeping them closed, then to return their hand to the starting position when prompted by the experimenter. The landing position of the participant’s finger was recorded by the experimenter to the nearest 0.5 cm, with positive values representing errors to the right of the target and negative values errors to the left.

During the open-loop pointing task in AR environment, participants wore the headset and, similarly to the real open-loop task, were verbally instructed (through the headset), to point six times towards the central blue target that was displayed in the head-mounted display. A black screen was displayed whenever the participant had to point, to prevent any visual feedback during

the movement. The distance between the landing position and the target was measured by the hand tracking device and recorded by the software. Both open-loop pointing tasks took approximately 1 min to complete. The sensorimotor shift was measured as the average distance between the landing position and the central target.

Landmark task

The landmark task is a perceptual judgment version of the line bisection task (Milner et al., 1992), often chosen to assess the visuospatial shift induced by PA (Colent et al., 2000; Michel et al., 2003; Michel & Cruz, 2015; Patané et al., 2016; Schintu et al., 2014, 2017). Participants were seated in front of a computer screen, positioned 35 cm from their eyes. In this two-alternative forced-choice task, they were presented with a series of pre-bisected white lines (350 mm x 3 mm) displayed centrally on a black screen and asked to judge whether the transector was closer to the left or the right end of the horizontal line. The lines were bisected at either 0 (true centre), 2, 4, 6, 8, or 10 mm towards the left or the right of the true centre, for a total of 66 trials. The task took about 4 min and was preceded by a short practice (~20 trials).

Participants answered by pressing a pedal with their left foot if they perceived the transector as being closer to the left end of the line and with their right foot if they perceived it closer to the right end of the line. The choice of collecting answers via pedals was made to limit the use of the right hand and therefore the de-adaptation process. Each pre-bisected line was displayed for a maximum of 5 s or until a response was made and was then replaced by a black-and-white patterned mask which stayed on the screen for 1 s before the next bisected line was displayed. Participants were instructed to respond as accurately as possible after having fully inspected the pre-bisected line, and to judge the transector position based on the line rather than based on their body midline. Presentation software (Neurobehavioral Systems, Inc., Berkeley, CA, www.neurobs.com) was used to generate the stimuli, record responses. For each participant, we considered how many times they answered “right” instead of “left,” and we plotted the percentage of these responses as a function of the transducer position. Using the software Matlab (Version: 9.1.0; R2016b), the point of subjective equality (PSE) was calculated for each block. Here, we computed the PSE as the point at which a participant answered “right” 50% of the times.

Statistical analysis

Statistical analyses were performed using Jasp (Version 0.17.1). To assess the immediate ARPA aftereffects, the average landing position of the AR-based open-loop task, indexing sensorimotor effects of ARPA and the PSE of the

landmark task, indexing the visuospatial effects, were submitted to two repeated measure analyses of the variance (ANOVA), with Time (Pre, Post 1) as a within-subjects factor and Direction (left ARPA, right ARPA) as a between-subjects factor. Since we were interested in comparing each post-adaptation session to the baseline for a given direction of ARPA, when significant the two-way interaction was followed by planned contrasts conducted independently for each ARPA direction, comparing pre-measurements with their respective post-measurements, as well as planned contrasts conducted between directions at each measurement point, thereby assessing the same measurements across different ARPA conditions. To account for multiple comparisons, we controlled the false discovery rate at 0.05 by adopting the Benjamini-Hochberg procedure. Accordingly, we reported the Benjamini-Hochberg adjusted p -values for each contrast. Estimates and 95% Confidence Intervals for Mean difference are reported for relevant contrasts. When the assumption of sphericity was violated, as indicated by Mauchly's test, Greenhouse-Geisser correction was applied. Hereafter, effect sizes are reported as partial eta squared.

To explore the time course of the sensorimotor shifts induced by ARPA, we run a repeated measures ANOVA having Time (Pre, Post 1–9) as a within-subjects factor, and Direction (left ARPA, right ARPA) as a between-subjects factor. A similar ANOVA was computed for the PSE values to evaluate the visuospatial shifts having Time (Pre, Post 1–8) as within-subjects factor and Direction (left ARPA, right ARPA) as a between-subjects factor.

To evaluate the generalization of the sensorimotor changes induced by ARPA when performing the open-loop pointing task within the AR scenario to when performing the task in the real environment, the pointing data from the two versions of the task were submitted to a repeated measures ANOVA with Task (augmented-reality, real pointing) and Time (Pre, Post 1, Post 9) as within-subjects factors and Direction (left ARPA, right ARPA) as a between-subjects factor.

Results

Immediate sensorimotor and visuospatial aftereffects of ARPA

Analysis of the AR-based open-loop pointing task revealed a significant main effect of Direction ($F_{1,46} = 105.55$, $p < .001$, $\eta^2 = .696$), indicating a general rightward shift for the left ARPA group (Estimate = 2.54, 95% CI = 1.67, 3.40) and a general leftward shift for the right ARPA group (Estimate = -3.67, 95% CI = -4.52, -2.87). The analysis also revealed a significant Direction by Time interaction ($F_{1,46} = 176.44$, $p < .001$, $\eta^2 = .793$). Contrasts revealed that for both groups the first post-adaptation session differed significantly from their respective baseline measurement, with a sensorimotor shift opposite to the deviation

of the visual field (left ARPA Estimate = 5.85, 95% CI = 4.49, 7.21; $t = 8.64$, $p = .002$; right ARPA Estimate = -6.87, 95% CI = -8.23, -5.51; $t = -10.14$, $p = .002$, see Pre vs Post 1 in Figure 3). While there was no significant difference between left and right directions prior to ARPA (Pre-ARPA Estimate = -0.16, 95% CI = -1.69, 1.35; $t = 0.20$, $p > .25$), this difference became significant after ARPA (Post-ARPA Estimate = 12.56, 95% CI = 11.03, 14.01; $t = 16.03$, $p = .005$, see Left vs Right in Figure 3). Analysis of PSE revealed a main effect of Direction ($F_{1,46} = 4.33$, $p = .043$, $\eta^2 = .086$), indicating an overall higher leftward bias in the right ARPA group (Estimate = -1.32 95% CI = -2.19, -0.44) as compared to the left ARPA group (Estimate = -0.04 95% CI = -0.91, .86), and a Direction by Time interaction ($F_{1,46} = 8.04$, $p < .001$, $\eta^2 = .149$). Contrasts indicated a significant, rightward shift in the midline judgment immediately after ARPA as compared to before, for the left ARPA group (Estimate = 1.05, 95% CI = 0.24, 1.85; $t = 2.60$, $p = .024$, see Pre vs Post 1 in Figure 4), while no significant shift was found for the right ARPA group (Estimate = -0.56, 95% CI = -1.37, 0.40; $t = 1.40$, $p > .25$). Contrasts also demonstrated that, prior to ARPA, there was no difference between the left and right groups (Pre-ARPA Estimate = 0.48, 95% CI = -0.87, 1.83; $t = 0.71$, $p > .25$). However, such a difference was found to be significant after ARPA (Post-ARPA Estimate = 2.08, 95% CI = 1.31, 3.43; $t = 3.08$, $p = .015$; see Left vs. Right in Figure 4). Taken together, these results show an immediate sensorimotor shift after a single session of either left or right ARPA, but an immediate visuospatial shift solely following a single session of left ARPA.

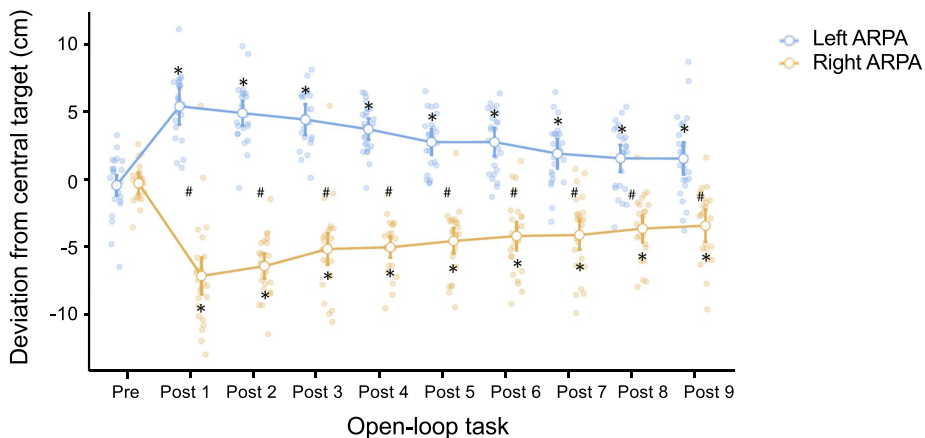


Figure 3. Augmented reality (AR) based open-loop pointing error (sensorimotor aftereffect) before and after an augmented reality-based adaptation to leftward (in blue) and rightward-deviating (in orange) ARPA. Zero represents no deviation from the central target. Negative and positive values represent errors to the left and right of the target. Error bars represent 95% CI. Filled dots represent observed cases. Asterisks indicate significant differences from baseline.

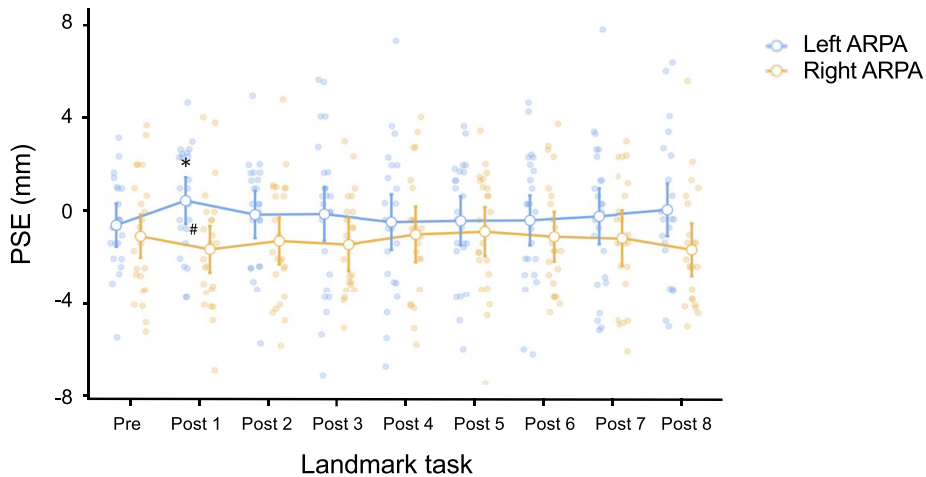


Figure 4. Point of subjective equality judgment (visuospatial aftereffect) task before and after augmented reality-based adaptation to leftward- (orange) and rightward-deviating prisms (in blue). Zero represents the line's true centre. Positive and negative values represent respectively judgments to the right and left of the line's true centre. Error bars represent 95% CI. Filled dots represent observed cases. Asterisks indicate significant differences from Pre-adaptation.

The time-course of ARPA

Concerning the AR open-loop task, the analysis revealed a significant main effect of Deviation ($F_{1,46} = 135.94, p < .001, \eta^2 = .747$) indicating a general rightward pointing for the left ARPA group (Estimate = 2.53, 95% CI = 1.73, 3.32) and a general leftward pointing for the right ARPA group (Estimate = -3.97, 95% CI = -4.76, -3.17), along with a significant Direction by Time interaction ($F_{4.32,197.89} = 45.26, p < .001, \eta^2 = .496$). Contrasts indicated that for both groups (left ARPA and right ARPA) all nine post-adaptation sessions differed significantly from baseline with a rightward shift in pointing values for the left experimental ARPA group (all $ps < .001$, all $ts > 3.11$) and a leftward shift in pointing values for the right control ARPA group (all $ps < .001$, all $ts > -4.52$; see Figure 3). Moreover, while the pointing values did not show a significant difference between left and right before ARPA ($t = -0.21, p > .25$), they became significant in all post-adaptation sessions (all $ps < .005$, all $t > 5.09$).

Concerning the landmark task, analysis of PSE did not reveal any significant main effect or interaction (all $Fs < 2.96, p > .092$). Following the reviewer's suggestion, we conducted a series of independent t -tests at each session on the delta values (post-pre) to determine whether, when accounting for the initial baseline, ARPA induces a differential visuospatial shift based on the direction. The analysis revealed a significant difference between left and right ARPA in the first delta values ($t = 2.85, p = .007$), indicating a visuospatial aftereffect that is evident only in the immediate aftermath of adaptation. To assess whether there was any correlation between an individual participant's shift in PSE and

the amplitude of their sensorimotor aftereffect (shift in open-loop pointing error), eight (one per post-adaptation session) independent correlations (Pearson's or Spearman's tests were used depending on normality assumption as assessed by Shapiro test) were computed between the absolute values of the two shifts. None of these correlations was significant (all $p > .144$).

Overall, these results show that a rightward and a leftward AR adaptation induce a leftward and a rightward sensorimotor aftereffect respectively, lasting at least 40 min.

Generalization from augmented reality to the real environment

The analysis on AR and real pointing errors showed a significant main effect of Direction ($F_{1,46} = 132.28, p < .001, \eta^2 = .742$) indicating an overall rightward shift for the left ARPA group (Estimate = 1.85, 95% CI = 1.21, 2.50) and an overall leftward shift for the right ARPA (Estimate = -3.31, 95% CI = -3.95, -2.67) group, along with a significant Direction by Time interaction ($F_{2,92} = 211.59, p < .001, \eta^2 = .821$). Contrasts indicated that, as compared to baseline, the pointing landing positions in both groups significantly shifted rightward in left ARPA group and leftward in the right ARPA group at both Post 1 and Post 9 (all $p < .001$: Left ARPA contrast Pre vs Post 1 Estimate = 5.79, 95% CI = 4.95, 6.63, $t = 13.70, p < .001$; Left ARPA Pre vs Post 9 Estimate = 1.51, 95% CI = 0.71, 2.35, $t = 3.59, p < .001$; right ARPA contrast Pre vs Post 1 = -6.321, 95% CI = -7.09, -5.38, $t = 14.66, p < .001$; right ARPA Pre vs Post 9 Estimate = -2.10, 95% CI = -2.94, -1.26, $t = -4.96, p < .001$). Furthermore, while the two groups did not differ before ARPA ($p > .25$), significant differences between Left and Right Direction emerged at Post 1 and Post 9 ($p < .001$). No other main effect or interaction were significant. The lack of a significant Time by Task interaction suggests that the amount of adaption as quantified by the two versions of the open-loop task did not differ. However, canonical statistics prevent from drawing any inference from the absence of a significant difference. To overcome this limitation, we complemented our analysis by performing Bayesian t-tests. To assess whether these non-significant results provided evidence for comparable sensorimotor shifts between the two tasks, we used the Bayesian factor comparing the null hypothesis against its complementary hypothesis (Kaplan & Depaoli, 2013; Patané et al., 2021, 2025; Rouder et al., 2009), reporting the inverse Bayes factor (BF01) if the Bayes factor implies support for the null hypothesis. For the left ARPA group, the Bayesian analysis favoured the null hypothesis according to which the amount of sensorimotor shift between the two open-loop tasks did not differ at both Post 1 (BF01 = 2.933), and Post 9 (BF01 = 2.24). Similarly, for the right control ARPA group, Bayesian factors supported the null hypothesis of similar sensorimotor shifts in the two tasks at Post 1 (BF01 = 3.184) and Post 9 (BF01 = 4.51). These complementary findings support our predictions about the generalization from AR to the real

environment and indicate similar sensorimotor shifts following adaptation to a leftward or rightward virtual prism when quantified using either an AR or a real open-loop pointing task. Notably, since such sensorimotor aftereffects are not only evident immediately after adoption (Post 1) but also at the very end of the experiment (Post 9), we can thus conclude that ARPA sensorimotor aftereffect lasts at least 40 min (see Figure 3).

Finally, we explored the association between the two sensorimotor measures by testing whether there was any correlation between an individual open-loop pointing performance in the AR version and in the real version of the task. Pearson correlations on the absolute values showed a positive association at Pre ($r = .424$, $p = .003$), Post 1 ($r = 0.430$, $p = 0.002$), and Post 9 ($r = 0.424$, $p = .003$).

To sum up, ARPA induces a sensorimotor aftereffect, which is not only present in a VR scenario where the adaptation took place but also transfers – in full – from the virtual settings to the real environment (Figure 5).

Discussion

The study aimed to investigate the effectiveness of adaptation to laterally displaced vision in AR environment in inducing both sensorimotor and visuospatial aftereffects, like those observed following the classic PA procedure. Specifically, we first examined whether ARPA induces an immediate sensorimotor and visuospatial shift. Then, we proceeded to identify the temporal dynamics of these effects and tested whether those could generalize into the real environment. As expected, results revealed that both right and left ARPA induced

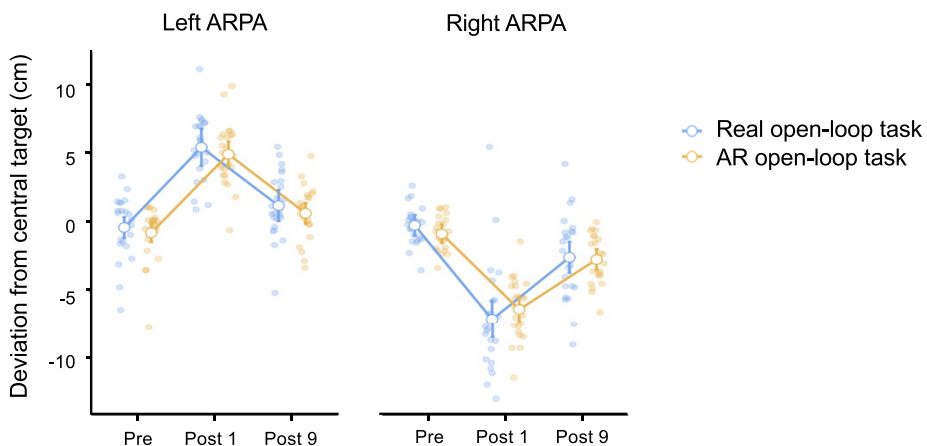


Figure 5. Real open-loop pointing error (blue) and augmented reality (AR)-based open-loop pointing error (yellow) for the left-deviating prisms group (left panel) and right-deviating prisms group (right panel), before (Pre), right after (Post 1) and after 40 min (Post 9). Zero represents no deviation from the central target. Negative and positive values represent errors to the left and right of the target, respectively. Error bars represent 95% CI of the mean. Filled dots represent observed cases.

immediate sensorimotor aftereffects and that only the left ARPA produced a rightward visuospatial aftereffect. Furthermore, the sensorimotor aftereffect induced by ARPA, but not the visuospatial shift, was long-lasting and generalized from the AR setting to the real environment.

This is the first study reporting that adaptation to a lateral displacement conducted in AR setting results in a significant and long-lasting sensorimotor aftereffect. The direction of this aftereffect, similar to the classical PA, depends on the direction of the lateral deviation induced by ARPA. Building upon a previous study by Schintu et al. (2014) investigating the temporal dynamics of classical PA, the current findings show that ARPA-induced sensorimotor aftereffects also persist for up to 40 min.

This work also presents a novel finding on the immediate visuospatial aftereffect induced by left ARPA, which is similar to the one induced by classical PA. Our results partially align with a previous study by Gammeri et al. (2018) where healthy individuals were adapted to prisms in a virtual environment, but no significant aftereffect was observed in the perceptual version of the line bisection task. However, there are noteworthy differences between Gammeri et al.'s (2018) study and our present work. The primary distinction lies in the type of test exhibiting a significant change in visuospatial performance after adaptation. In our study, we quantified the visuospatial shift using the landmark task, which revealed a significant effect immediately after ARPA. In contrast, Gammeri et al. (2018) employed both the landmark task and manual line bisection task but found a significant change in performance solely in the manual line bisection task, at odds with previous reports suggesting that the landmark task is more sensitive to traditional PA (Fortis et al., 2011; Michel et al., 2003; Striemer & Danckert, 2010). Most notably, the absence of an effect at the landmark task could be attributed to the fact that the authors adapted healthy individuals to a rightward displacement (right PA), a direction typically used to ameliorate neglect patients' behaviour but that, as shown by most studies, hardly modulate visuospatial cognition in intact brains (Bultitude et al., 2013; Colent et al., 2000; Loftus et al., 2009; Michel & Cruz, 2015; Patané et al., 2016; Reed & Dassonville, 2014; Schintu et al., 2014, 2018, 2020, 2022). The direction of the displacement, however, would not account for the significant effect at the manual line bisection task. As reported by Gammeri et al.'s (2018), in their study participants were divided into 4 groups according to the maximum strength of the PA displacement, 0°, 10°, 20°, and 30°, and solely the group who had the maximum visual shift of 30° showed a significant change at the manual line bisection task. It has been shown that increasing the power for the deviation increases the magnitude of the induced aftereffects (Michel & Cruz, 2015). However, the fact that a displacement of 30° is not achievable with typical wedge lenses (i.e., classical PA) and the maximum deviation used in PA is 15°, prevents a direct comparison between the two techniques. Crucially, the main difference between landmark and manual line bisection is the activation of a motor

programme in the latter, and prior evidence in patients supports a distinction between motor and purely perceptual PA aftereffects (Ferber et al., 2003; Gammeri et al., 2023; Striemer & Danckert, 2010). It is therefore possible that such strong, 30°, rightward deviation affected the motor-based component present in the manual line bisection task also in healthy individuals but still had no effect on the perceptual component assessed by the landmark task (Milner et al., 1992). To comprehensively understand the potential dissociation between perceptual and motor aspects of bisection performance, it is important to address a significant gap in the existing literature. Specifically, critical comparisons are lacking for the aftereffects resulting from both canonical and virtual protocols of PA with varying degrees of lateral displacement in rightward and leftward directions. A systematic study investigating the impact of both PA direction and magnitude on spatial processing in healthy individuals would be valuable to develop more evidence based computational and theoretical models that can be evaluated in intact brains and applied then to neglect patients.

Overall, our findings confirm that ARPA does induce both visuomotor and visuospatial aftereffects. However, contrary to our prediction, the visuospatial change induced by left ARPA did not last as long as the one observed with classical PA (Schintu et al., 2014). While ARPA induced an immediate post-adaptation effect, the visuospatial changes associated with classical PA persisted, albeit with some fluctuations, for up to 40 min. This discrepancy warrants further investigation into the underlying mechanisms. Interestingly, this lack of prolonged modulation in visuospatial aftereffects cannot be attributed to an ineffective adaptation procedure. Participants' sensorimotor aftereffect was, in fact, significant until the end of the experiment (i.e., the last post-session) and, by visual inspection, the ARPA-induced sensorimotor aftereffects observed here are similar to those reported in Schintu et al. (2014), where the dynamics of PA-induced aftereffects were investigated similarly.

Considering that ARPA produces a pattern of sensorimotor aftereffects similar to the one repeatedly observed after classical PA, it is worth discussing that the robustness of visuospatial aftereffects induced by classical PA has recently been questioned (McIntosh et al., 2023). While a meta-analysis suggested that an extended period of adaptation to 15° leftward prisms should robustly induce this visuospatial aftereffect (McIntosh et al., 2019), a subsequent pre-registered report failed to replicate this effect on a larger sample and proposed that the effect size estimated from prior literature might be over-optimistic (McIntosh et al., 2023). Although the authors did not rule out the possibility that leftward PA might induce higher-level visuospatial aftereffects in healthy individuals, they warned that caution should be taken when interpreting the visuospatial aftereffects. However, some differences in methodologies should be highlighted here. McIntosh et al. utilized a "sham-adaptation" control, which does not correspond to our approach of using the rightward ARPA group. Unlike our version of the landmark task we employed, the lines

in their task were translocated by 0.5, 1, 2, 4, or 8 mm to the left or right of the centre, with 16 lines for each condition. Some of these 16 lines were centred on the screen, whereas some lines were shifted by 3, 6, and 9 mm to both the left and right. The potentially different effects on visuospatial performance of these different transectors and the lateral shifts of the line remain unknown. While we acknowledge the necessity of a larger sample size in our ARPA study to replicate our results, research should address this critical issue regarding the various types of stimuli utilized in the landmark task to measure the PA-induced visuospatial aftereffects. Because visuospatial aftereffects of ARPA are a consequence of sensorimotor adaptation, we might expect a correlation between the magnitude of the sensorimotor aftereffects and visuospatial aftereffects. Nevertheless, while we did not find any evidence for such a relationship this null result aligns with previous findings, as highlighted by McIntosh et al. (2023; see Michel, 2016, for a review).

Interestingly, we observed that the amount of sensorimotor adaptation induced by ARPA is similar across virtual (adaptation context) and real environments (non-adaptation context). In addition, this held true both immediately (Post 1) as well as 40 min after ARPA (Post 9). These findings underscore the capability of ARPA device and method to generalize sensorimotor aftereffects beyond the virtual environment in which the adaptation took place. It is important to note that in our study the environment was augmented reality, not fully virtual, which may account for the transfer of effects to the real world. However, this should be seen as an advantage of AR-based techniques, as their hybrid nature (part real, part virtual) may enhance the likelihood of adaptation transferring to the real environment.

Although the future implications of ARPA in rehabilitation must be approached with caution, due to its short-lived effect on visuospatial performance, we believe that the augmented or virtual version of the classical PA procedure may still hold potential for clinical applications. Current literature suggests that VR-based technology offers innovative and promising opportunities for rehabilitation, with its effectiveness potentially depending more on the quality of the simulation itself rather than the degree of immersion. Nevertheless, further randomized controlled trials are needed to establish its clinical applications (Salatino et al., 2023). In fact, after years of enthusiasm for PA's effectiveness in ameliorating neglect patients' symptoms, recent meta-analyses (Li et al., 2021; Qiu et al., 2021) have raised significant concerns regarding its efficacy in clinical populations by suggesting that the positive outcomes observed in controlled experimental settings do not consistently translate into clinical efficacy for patients with neglect. A subsequent meta-analysis further questioned its effectiveness, (Székely et al., 2023) concluding that PA, in the short term, has no effects on neglect patients' performance, as assessed by cancellation tasks (notably, this conclusion overlooked findings from line bisection and other assessments). This implication, however, has been

challenged by Lunven et al. (2023), who suggest that PA effectiveness in neglect patients may depend on the connectional anatomy of their lesion, in addition to other factors such as demographics and clinical characteristics, as well as preserved intra and interhemispheric connectivity (Kaufmann et al., 2024). It has been additionally put forward that PA may be most effective in individuals with attention and motor intention deficits due to frontal rather than parietal lesions (Carter & Barrett, 2023; Scheffels et al., 2022). These findings indicate that several clinical and patient-related factors should be considered before implementing this procedure. Although this intervention is not recommended in most clinical guidelines, it is noteworthy that the 2016 AHA guidelines for adult stroke rehabilitation (Winstein et al., 2016) recommend considering PA as a treatment option. This recommendation is derived from assessments of established non-pharmacological interventions. Moreover, literature is now re-evaluating the efficacy of PA, underscoring the importance of using all available rehabilitation tools for this challenging condition (Hildebrandt et al., 2024). Clinicians find PA particularly intriguing because it is non-invasive, bypasses neglect patients' lack of awareness, requires minimal training, and its application takes only a few minutes. Our ARPA approach not only is similar to the classical PA for the aforementioned characteristics, but it is even more flexible. Once tested for its efficacy on neglect patients, ARPA could, indeed allow for easy remote handling by an operator while still maintaining the rigour of experimental classical PA. Since patients can use this device autonomously, even outside their rehabilitation sessions and potentially independently at home, ARPA may enable patients to actively participate in their own health and recovery process for longer treatment periods. Meanwhile, healthcare professionals could maintain control over critical parameters and treatment settings to ensure its effectiveness. ARPA could thus represent a promising clinical tool for both neglect assessment and rehabilitation. It could be used to observe and measure behavioural patterns that may not be evident during standard testing, but become clear in more natural, ecological settings, especially as patients interact with their environment. This ecological approach allows interventions to be customized to each patient's specific strengths and weaknesses, potentially unlocking unexplored opportunities in diagnosis and treatment. It may therefore be beneficial to involve not only patients but also caregivers and clinicians in choosing the most suitable, personalized, ecological treatment plan.

It is worth noting that during our study, we did not observe any adverse effects such as headaches or motion sickness among participants. The AR headset and settings were well-tolerated by all individuals, positioning our device as a promising candidate for inclusion in rehabilitative protocols. While ARPA shares many advantages with classical PA, such as non-invasiveness and minimal training requirements, its integration into clinical practice will surely depend on its ability to produce lasting functional improvements. Indeed, the next step would be to test this approach with patients to assess

its validity as a rehabilitation tool, and we hope that more ARPA-like techniques will be explored for rehabilitation purposes, allowing the creation of better-tailored rehabilitation protocols not only for spatial neglect but also for other conditions such as postural balance (Hugues et al., 2015; Tilikete et al., 2001), chronic pain (Christophe et al., 2016), Parkinson's disease (Bultitude et al., 2013), or developmental dyslexia (Conte et al., 2024), as studies have shown they benefit from prism adaptation.

These results suggest that ARPA could represent a critical innovation, but it is worth noting that it is an experimental intervention and, as such, requires validation before any clinical application. Future work should elucidate the mechanisms underlying ARPA-related effects, determine its potential and limitations as a therapeutic tool, and validate its possible clinical relevance. This evaluation should extend beyond the landmark task for measuring perceptual bias to include a broader assessment that effectively gauges its impact on visuospatial attention. Clinical studies should encompass both short- and long-term investigations to determine the durability of sensorimotor and visuospatial aftereffects and explore how ARPA can be integrated into existing rehabilitation protocols and personalized treatment plans. A critical aspect of this research will be evaluating the impact of ARPA on functional outcomes, such as daily activities, mobility, and quality of life, to ascertain its practical value in clinical settings. Results from this line of research could open the possibility of using ARPA as a complement to traditional wedge prisms, exploiting the advantages and flexibility of this technology to implement customized rehabilitative protocols for neglect patients.

In conclusion, this is the first study investigating the dynamic of the after-effects induced by adaptation to laterally displaced vision in an AR setting. Given the contrasting findings in the literature regarding PA efficacy on neglect symptoms, our study should be considered an important step in demonstrating ARPA potential in healthy individuals. These findings suggest that while ARPA holds the potential for inducing sensorimotor aftereffects, its short-lived visuospatial effect following a single PA session highlights the need for cautious interpretation and further studies to evaluate its applicability in clinical settings.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability

Data will be made available on request.

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